

(REFERENCE SPECIFICATION SECTION 48 19 23.08 26)

- TYPE: CLASS II GSI POWER TRANSFORMER
- COOLING CLASS: ONAN/ONAF
- KVA RATING: 164,000 / 218,666
- VOLTAGE RATINGS: 500,000 GRDY (H) - 13,200 DELTA (X) 13,200 DELTA (Y)
- BIL LEVELS, KV: 1,425 (H) / 110 (X & Y) / 150 (HO)
- PHASING: SINGLE-PHASE
- IMPEDANCE: 17.7% @ 82,000 KVA (H-X & H-Y),
- SECONDARY WINDING (H) DETC: 5 TAPS; 3 @ +2.5%, 1 @ -2.5% OF NOMINAL 500 KV TAP
- TEMPERATURE RISE: 65 °C
- OIL PRESERVATION SYSTEM: NITROGEN BLANKET (PREFERRED) OR CONSTANT OIL PRESSURE SYSTEM
- MAXIMUM SOUND LEVEL (ONAF): 80 DB
- MAXIMUM NO-LOAD LOSSES: 165 KW
- MAXIMUM LOAD LOSSES: 314 KW
- FREQUENCY: 60 HZ
- MAXIMUM WEIGHT: 596,500 POUNDS
- EXISTING OIL QUANTITY: 15,539 GALLONS

- MAN-HOLE: 24-INCH DIAMETER (MINIMUM TWO)
- GASKETS: NITRILE NBR (BUNA-N), DURO 50-60, 25% COMPRESSION, WITH THE EXCEPTION THAT THE LOW-VOLTAGE BUSHING GASKETS SHALL BE FLUORELASTOMERIC RUBER.
- O-RING GASKETS SHALL NOT BE USED FOR GASKET GROOVES OF RECTANGULAR CROSS-SECTION.
- VALVES: STAINLESS STEEL BALL-TYPE (RADIATORS MAY USE BUTTERFLY-TYPE)
- PERSONAL FALL ARREST SYSTEM: THE TOP OF THE TRANSFORMER TANK SHALL BE COVERED WITH A NON-SLIP MATERIAL, THE SAME COLOR AS THE TANK WALLS. A FALL PROTECTION SYSTEM, CONSISTING OF A SINGLE ANCHOR POST, SHALL BE PROVIDED.
- BASE: DESIGNED FOR PAD MOUNTING WITH SEISMIC WELDING TO STEEL PLATES.
- LIFTING EYES: DESIGN LOADS FOR EACH EYE SHALL BE TWO TIMES THE ACTUAL LOAD.
- SEISMIC: SEISMIC CALCULATIONS REQUIRED FOR MAIN TANK AND ALL MAJOR COMPONENTS.
- CORE: LAMINATIONS MAXIMUM THICKNESS OF 0.270MM, COLD-ROLLED, GRAIN-ORIENTED STEEL. MAXIMUM DESIGN FLUX DENSITY OF 1.75 TESLA. STEP-LAP CONSTRUCTION.
- RADIATORS: HOT-DIP GALVANIZED.
- COOLING FANS: HORIZONTAL-DRAFT
- PROVIDE LOW-VOLTAGE HOUSINGS TO CONNECT THE TRANSFORMER TO THE ISOPHASE BUS COVER
- THE GOVERNMENT WILL USE THE EXISTING COPPER BUS BARS THAT CONNECT THE LV BUSHING TO THE ISOPHASE BUS WHEN PERMENTLY INSTALLING THE SPARE AT A PROJECT

- WTI AND OTI GAUGES WITH RTDS - HARDENED FOR HARSH WEATHER CONDITIONS
- DIRECTED-FLOW PRESSURE RELIEF DEVICE WITH METAL DISCHARGE PIPING
- TRANSFORMER EMBEDDED FIBER-OPTIC WINDING TEMPERATURE MEASUREMENT PROBES

- SECONDARY WINDING (H1): 500 KV CLASS, MINIMUM 1,500 KV BIL, 1,600 AMP, OIL-FILLED BUSHING
HITACHI TYPE O+ C T550W2000VV
(TWO 1200.5 MRCT)
- PRIMARY WINDING: 25 KV CLASS, MINIMUM 150 KV BIL, 10,000 AMP, PORCELAIN CONSTRUCTION,
BULK-TYPE, LOW-CORONA, LIGHT CONTAMINATION, 125 °C TEMPERATURE RISE RATING
HITACHI TYPE RJ T025B1000DS
(BUSHING CURRENT TRANSFORMERS AS REQUIRED FOR THE WINDING TEMPERATURE
INDICATING EQUIPMENT)
- NEUTRAL (H0): 25 KV CLASS, MINIMUM 150 KV BIL, 1,200 AMP
HITACHI TYPE O+ C T025W0412AT
- CORE GROUNDS: 2.5 KV CLASS, 45 KV BIL, 100 AMP

· TYPE: NAPHTHENIC BASE - ASTM D 3487 TYPE II; 0.3% BY MASS OF OXIDATION INHIBITOR

- CABINET MATERIAL: WEATHER-PROOF TYPE 304 STAINLESS STEEL WITH LOCKING PROVISIONS. WINDOWS FOR VIEWING THE FUTURE TEMPERATURE MONITOR AND ANNUNCIATOR IN CONTROL CABINET. SOLAR SHIELDING ON TOP, FRONT AND SIDE PANELS OF ENCLOSURES.
- GOVERNMENT-FURNISHED POWER: CONTROL POWER - 125 V DC; ANNUNCIATION POWER - 125 V DC; COOLING FANS - 480 V AC, 3-WIRE, 3-PHASE, 60 HZ
- WIRING: XLPE; 3-PHASE POWER CONDUCTORS COLOR CODED: PHASE A - BROWN; PHASE B - ORANGE; PHASE C - YELLOW; GROUNDED NEUTRAL - WHITE; INSULATED GROUND - GREEN/YELLOW
- CONDUIT: ALL WIRING TO BE RUN IN CONDUIT. WIRE RUNS THROUGH TANK REINFORCING CHANNELS NOT ALLOWED. ALL CONDUIT THREADED JOINTS SHALL HAVE A CONDUCTIVE JOINT COMPOUND APPLIED. OUTLET AND JUNCTION BOXES SHALL BE MALLEABLE CAST IRON WITH CAST COVERS AND LIQUID-TIGHT HUBS. CONDUIT FILL SHALL BE IN ACCORDANCE WITH NFPA 70.
- TERMINAL BLOCKS: 600 V, 30 A MINIMUM, 90 °C RATED TERMINATIONS.
- AUXILIARY RELAYS:
 - RATED FOR 125 V DC OR 120 V AC; MEETS REQUIREMENTS OF NEMA ICS5; 10 A CONTACTS
 - SINGLE-CONTACT INDUCTIVE LOAD INTERRUPTING CAPACITY: 125 V DC CIRCUITS - 1.0 A; 120 V AC CIRCUITS - 6.0 A
- SWITCHES:
 - RATED FOR 600 V AC OR 250 V DC
 - SINGLE-BREAK INDUCTIVE LOAD INTERRUPTING RATING: 125 V DC CIRCUITS - 1.5 A; 120 V AC CIRCUITS - 10 A
- MOLDED-CASE CIRCUIT BREAKERS:
 - 480 V AC CIRCUITS - RATED 600 V AC AND 18,000 A INTERRUPTING AT 480 VAC
 - 120 V AC CIRCUITS - RATED 250 V DC AND 120/240 OR 240 V AC, AND 10,000 A INTERRUPTING
 - 125 AND 250 V DC CIRCUITS - RATED 125/250 OR 250 V DC, AND 10,000 A INTERRUPTING
- SPECIAL FEATURES:
 - ALL CIRCUITS, WIRING AND EQUIPMENT RATED GREATER THAN 120 V AC TO BE MOUNTED IN POWER CABINET.
 - 120 V AC, 1,000 LUMEN LED TYPE A19 LAMPS WITH DOOR-ACTUATED SWITCH.
 - 120 V AC, 20 A HOSPITAL GRADE GFCI RECEPTACLE MOUNTED INSIDE THE CONTROL CABINET.
 - PTC RESISTOR DESIGN ENCLOSURE HEATERS WITH INTEGRAL FANS, AND INDUSTRIAL GRADE HUMIDISTAT / THERMOSTAT CONTROL.
 - FOLD-DOWN MAINTENANCE TRAY ON INSIDE OF CONTROL CABINET DOOR.
- NAMEPLATES: CAST BRONZE OR STAINLESS STEEL; IEEE C57.12.00, TABLE 7, INCLUDING MANUFACTURING DATE, PLANT LOCATION, GOVERNMENT CONTRACT NUMBER, PRESSURE AND VACUUM THE TANK WILL WITHSTAND, DRY WEIGHT OF THE INSULATION, AND CALCULATED X/R RATIO. AN ADDITIONAL DIAGRAM NAMEPLATE FOR SLING LOCATIONS AND LIFT POSITIONS SHALL ALSO BE PROVIDED. EQUIPMENT NAMEPLATES SHALL BE PROVIDED FOR THE HIGH-VOLTAGE BUSHINGS AND LOW-VOLTAGE BUSHINGS MADE OF STAINLESS STEEL WITH BLACK LETTERING. DEVICE NAMEPLATES WITHIN THE CONTROL AND POWER CABINETS SHALL BE ANODIZED ALUMINUM OR STAINLESS STEEL WITH ENGRAVED BLACK LETTERING.

- ASSEMBLY: TRANSFORMER TO BE FULLY ASSEMBLED, WITH REMOVABLE COMPONENTS MATCH-MARKED FOR EASE OF REASSEMBLY.
- EXTERIOR PAINT: PRIMER COAT, EPOXY BASE COAT, POLYURETHANE TOP COAT - MUNSELL 5BG 7/0.4 HIGH-GLOSS (ANSI NO. 70 GRAY). CORROSION RESISTANT SURFACES, AND STAINLESS STEEL CABINETS, SHALL REMAIN UNPAINTED. COVER TO RECEIVE A NON-SLIP COATING.
- INTERIOR PAINT: PRIMER COAT, OIL-RESISTANT FINISH COAT - WHITE. ALLOW TO CURE FOR ONE WEEK PRIOR TO OIL FILLING.
- INSPECTIONS: GOVERNMENT INSPECTIONS PERFORMED DURING
 - (1) DESIGN REVIEW MEETING
 - (2) AFTER COMPLETION OF WINDINGS BUT PRIOR TO LANDING ON CORE
 - (3) COMPLETED CORE/COIL ASSEMBLY, BUT PRIOR TO TANKING (IF PROVIDING A CORE TYPE TRANSFORMER)
 - (4) COMPLETELY ASSEMBLED AND WITNESSING OF FACTORY ACCEPTANCE TESTS

- ALL 'ROUTINE' TESTS (IEEE C57.12.00, TABLE 17)
- TEMPERATURE RISE TESTS SHALL BE PERFORMED AT EACH COOLING STAGE
- LIGHTNING IMPULSE TESTS
- AUDIBLE SOUND LEVEL TESTS SHALL BE PERFORMED
- INDUCED VOLTAGE TEST ACCEPTANCE CRITERIA: 7,200 CYCLE ENHANCEMENT LEVEL: 500 PC
ONE HOUR LEVEL: 300 PC
- LEAKAGE REACTANCE (WITH ELECTRONIC DATA FILES)
- SWEEP FREQUENCY RESPONSE ANALYSIS (WITH ELECTRONIC DATA FILES)
 - CABLE INTEGRITY TEST REQUIRED
 - ONE SFRA TEST TO BE PERFORMED WITH THE TRANSFORMER FULLY ASSEMBLED AND OIL FILLED
 - ONE SFRA TEST TO BE PERFORMED WITH THE TRANSFORMER IN THE SHIPPING CONFIGURATION
- DE-ENERGIZED TAP CHANGER MECHANICAL OPERATIONAL TEST (5 COMPLETE CYCLES)
- ALL ELECTRICAL AND ELECTO-MECHANICAL DEVICES SHALL BE FUNCTIONALLY TESTED
- EACH HIGH-VOLTAGE AND HIGH-VOLTAGE NEUTRAL BUSHING SHALL UNDERGO CAPACITANCE AND POWER FACTOR TESTS DURING TRANSFORMER F.A.T.
- BUSHING CURRENT TRANSFORMER ROUTINE TESTS
- ANALOG TEMPERATURE GAGE FUNCTIONAL VERIFICATION

- (QTY 1): COMPLETE GASKET SET
- (QTY 1): SET OF GASKETS FOR TRANSFORMER REASSEMBLY
- (QTY 1): RELAY AND CONTACTOR OF EACH TYPE AND RATING
- (QTY 1): PRESSURE RELIEF DEVICE
- (QTY 1): LIQUID LEVEL INDICATOR
- (QTY 1): WINDING TEMPERATURE INDICATOR
- (QTY 1): OIL TEMPERATURE INDICATOR
- (QTY 1): BUSHING CURRENT TRANSFORMER OF EACH TYPE AND RATING
- (QTY 1): ONE RAPID PRESSURE RELAY AND SEAL IN RELAY
- (QTY 2): COOLING FAN COMPLETE WITH MOTOR, CABLES AND CONNECTORS
- (QTY 1): CRATED RADIATOR, COMPLETE WITH INLET AND OUTLET ISOLATION VALVES, VENT AND DRAIN VALVES
- (QTY 3): FIBER OPTIC EXTENSION CABLE THAT CONNECTS THE FEED-THROUGH WALL PLATE LEFT UNCONNECTED
- (QTY 1): FUSE OF EACH TYPE AND RATING
- (QTY 1): PRESSURE REGULATOR (IF SUPPLYING NITROGEN BLANKET TYPE OIL PRESERVATION SYSTEM)
- (QTY 1): BUCHHOLZ RELAY (IF SUPPLYING A COPS TYPE OIL PRESERVATION SYSTEM)
- (QTY 1): VALVE OF EACH SIZE AND TYPE



Diagram illustrating the correct method for grounding a core frame:

- Left Side (Correct Method):** Shows a cross-section of a core frame with two vertical cores. A protective cover is mounted on top. Grounding cables (#4 AWG) are connected to the core ground bushings on the cover. Labels include: GROUNDING CABLE (#4 AWG), CORE GROUND BUSHING, CORE FRAME GROUND BUSHING, TANK COVER, CORE, and CORE FRAME.
- Right Side (Incorrect Method):** Shows a similar setup but with the grounding cable connected to the core frame instead of the bushing. Labels include: GROUNDING CABLE (#4 AWG), TANK COVER, and CL.

E10 CORE AND CORE FRAME GROUNDING
SCALE: NTS



A9 EXISTING LV BUSHING BUS BARS
SCALE: NTS

1. THE TRANSFORMER LAYOUT AND LIMITING DIMENSIONS HAVE BEEN PROVIDED FOR THE MANUFACTURER'S USE IN DESIGNING THE TRANSFORMER FOR PROPER INTERCONNECTION TO THE EXISTING EQUIPMENT AND INTERFACES IN THE TRANSFORMER VAULT. THE LOCATIONS OF THE FOLLOWING TRANSFORMER EQUIPMENT ARE CONSIDERED CRITICAL TO ASSURE PROPER INTERCONNECTION:

- A. HIGH-VOLTAGE BUSHING LOCATION AND TERMINAL CONNECTION
- B. LOW-VOLTAGE BUSHINGS LOCATIONS AND TERMINAL CONNECTIONS
- C. HIGH-VOLTAGE NEUTRAL BUSHING LOCATION AND TERMINAL CONNECTION
- D. LOW-VOLTAGE BUSHING HOUSING DIMENSIONS AND LOCATION OF LOW-VOLTAGE BUS
- E. CONTROL AND POWER CABINETS
- F. ANCILLARY DEVICES
- G. JACKING & LIFTING FACILITIES
- H. STRUCTURAL BASE
- I. MAXIMUM OVERALL WIDTH
- J. MAXIMUM OVERALL LENGTH

2. THE DRAWING SHOWN IS BASED ON THE WESTINGHOUSE SHELL-FORM DRAWINGS WITH AN INERT-GAS OIL PRESERVATION SYSTEM. IF A CORE-FORM TRANSFORMER AND/OR A CONSTANT OIL PRESERVATION SYSTEM IS PROVIDED, THE PERMISSIBLE CLEARANCES IDENTIFIED MUST BE MET. THE COPS TANK MUST MAINTAIN A MINIMUM OF 105" CLEARANCE FROM THE OVERHEAD 500KV CONDUCTORS PER IEEE STD 1427. THERE SHALL BE NO MODIFICATIONS TO THE POWERHOUSE EQUIPMENT NEEDED TO FIT THE COPS TANK. IF CONSTANT OIL PRESERVATION SYSTEM IS PROVIDED, ALL ASSOCIATED EQUIPMENT REQUIRED IN SECTION 48 19 23.08 26 FOR A CONSTANT OIL PRESERVATION SYSTEM MUST BE INCLUDED, SUCH AS A MAINTENANCE-FREE DEHYDRATING BREATHER, AND BUCHHOLZ RELAY.

3. THE RADIATORS AS DEPICTED INDICATE THE PERMISSIBLE LOCATIONS, AND ARE NOT INTENDED TO INDICATE THE ACTUAL QUANTITY.

4. PROVIDE WEATHER PROOF LV HOUSINGS WITH ACCESS COVER THAT ARE EASY TO REMOVE AND ALLOWS ACCESS TO THE LV BUSHINGS AND THE FLEXIBLE CABLES THAT CONNECT TO THE ISOPHASE BUS. THE EXISTING COPPER BUS BARS WILL BE REUSED TO CONNECT THE BUSHING TO THE ISOPHASE BUS AND FLEXIBLE CONDUCTORS. SEE SECTION H1 AND A9.

5. MAXIMUM WEIGHT. 500,000 POUNDS

6. SEE RESOURCE DRAWINGS FOR PAD DETAILS.

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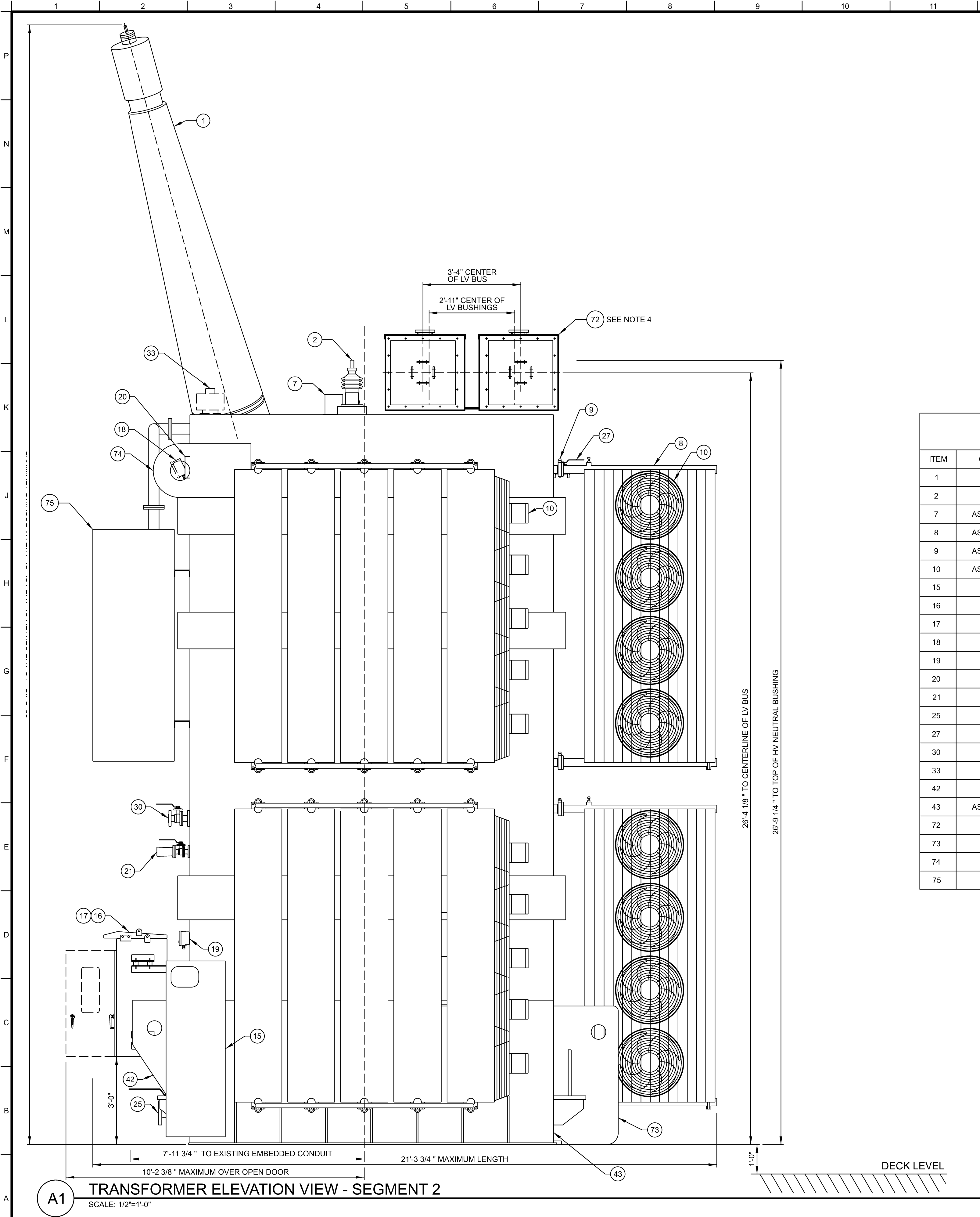
U.S. ARMY CORPS OF ENGINEERS WALLA WALLA DISTRICT 201 N 3RD AVE WALLA WALLA, WASHINGTON	DESIGNED BY: P. KLEINE, P.E.	ISSUE DATE: OCTOBER 2025
	DRAWN BY: W. J. KLEINE, P.E.	SOLICITATION NO.: WJZ-25RA003
	CHECKED BY:	CONTRACT NO.:
	D. FILER	
U.S. ARMY CORPS OF ENGINEERS HYDROELECTRIC DESIGN CENTER 333 SW 1ST AVE PORTLAND, OREGON	SUBMITTED BY: JORDAN S. FINKE, P.E.	FILE NUMBER: LGS-1-6-1E401
	APPROVED BY: ANSHU	PROJECT NO.: F-1014-1

LITTLE GOOSE LOCK AND DAM
LOWER SNAKE RIVER SPARE TRANSFORMERS
T1

GENERATOR STEP-UP TRANSFORMER
TOP VIEW



SHEET ID
VOLUME 1
E-101



GENERAL NOTES

1. THE TRANSFORMER LAYOUT AND LIMITING DIMENSIONS HAVE BEEN PROVIDED FOR THE MANUFACTURER'S USE IN DESIGNING THE TRANSFORMERS FOR PROPER INTERCONNECTION TO THE EXISTING EQUIPMENT AND INTERFACES IN THE TRANSFORMER VAULT. THE LOCATIONS OF THE FOLLOWING TRANSFORMER EQUIPMENT ARE CONSIDERED CRITICAL TO ASSURE PROPER INTERCONNECTION:
- A. HIGH-VOLTAGE BUSHING LOCATION AND TERMINAL CONNECTION
 - B. LOW-VOLTAGE BUSHING LOCATIONS AND TERMINAL CONNECTIONS
 - C. HIGH-VOLTAGE NEUTRAL BUSHING LOCATION AND TERMINAL CONNECTION
 - D. LOW-VOLTAGE BUSHING HOUSINGS DIMENSIONS AND LOCATIONS OF LOW-VOLTAGE BUS
 - E. CONTROL AND POWER CABINETS
 - F. ANCILLARY DEVICES
 - G. JACKING & LIFTING FACILITIES
 - H. STRUCTURAL BASE
 - I. MAXIMUM OVERALL LENGTH
 - J. LOCATION OF EXISTING EMBEDDED CONDUIT COMPARED TO THE CONTROL AND POWER CABINET
2. THE DRAWING SHOWN IS BASED ON THE WESTINGHOUSE SHELL-FORM DRAWINGS WITH AN INERT-GAS OIL PRESERVATION SYSTEM. IF A CORE-FORM TRANSFORMER AND/OR A CONSTANT OIL PRESERVATION SYSTEM IS PROVIDED, THE DIMENSIONAL REQUIREMENTS IDENTIFIED MUST BE MET. THE COPS TANK MUST MAINTAIN A MINIMUM OF 105" CLEARANCE FROM THE OVERHEAD 500KV CONDUCTORS PER IEEE STD 1427. THERE SHALL BE NO MODIFICATIONS TO THE POWERHOUSE EQUIPMENT NEEDED TO FIT THE COPS TANK. IF CONSTANT OIL PRESERVATION SYSTEM IS PROVIDED, ALL ASSOCIATED EQUIPMENT REQUIRED IN SECTION 48 19 23 08 26 FOR A CONSTANT OIL PRESERVATION SYSTEM MUST BE INCLUDED, SUCH AS A MAINTENANCE-FREE DEHYDRATING BREATHING, AND BUCHHOLZ RELAY.
3. THE RADIATORS AS DEPICTED INDICATE THE PERMISSIBLE LOCATIONS, AND ARE NOT INTENDED TO INDICATE THE ACTUAL QUANTITY.
4. PROVIDE LOW-VOLTAGE BUSHING HOUSINGS AS INDICATED. SEE SECTION H1 ON SHEET E-101 FOR DIMENSIONAL REQUIREMENTS AND DETAILS.
5. MAXIMUM WEIGHT: 696,500 POUNDS
6. SEE RESOURCE DRAWINGS FOR PAD DETAILS.

BILL OF MATERIALS

ITEM	QTY	SEGMENT	DESCRIPTION	EXAMPLE	MODEL NUMBER
1	1	COVER	HIGH-VOLTAGE (H1) BUSHING	HITACHI	TYPE O + C T550W2000VV
2	1	COVER	HIGH-VOLTAGE NEUTRAL (H0) BUSHING	HITACHI	TYPE O + C T025W0412AT
7	AS DES.	COVER	CORE & CORE FRAME GROUND BUSHINGS		
8	AS DES.	1, 2, 4	RADIATORS (TYPICAL)		
9	AS DES.	1, 2, 4	RADIATOR ISOLATION VALVES		
10	AS DES.	1, 2, 4	HORIZONTAL DRAFT COOLING FANS	KRENZ-VENT	
15	1	1	DUAL TANK NITROGEN REGULATOR CABINET	ABB	RND2
16	1	3	CONTROL CABINET		
17	1	3	POWER CABINET		
18	1	3	OIL LEVEL INDICATOR	QUALITROL	042 SERIES
19	3	3	ANALOG TEMPERATURE INDICATORS	QUALITROL	TR6000 SERIES
20	5	3	THERMOWELLS (QTY. 3 - STANDARD, QTY. 2 - HEATED)		
21	1	3	RAPID PRESSURE RISE RELAY WITH 1-1/2-INCH ISOLATION VALVE	QUALITROL/MILWAUKEE VALVE	900-FLA & F20 SERIES, SS, CLASS 150
25	1	3	3-INCH MAIN OIL DRAIN VALVE	MILWAUKEE VALVE	F20 SERIES, SS, CLASS 150
27	1	1	1.5-INCH UPPER FILTER VALVE	MILWAUKEE VALVE	F20 SERIES, SS, CLASS 150
30	2	3	2-INCH GAS-IN-OIL MONITOR INTERFACE VALVES	MILWAUKEE VALVE	F20 SERIES, SS, CLASS 150
33	1	4, COVER	PRESSURE RELIEF DEVICE WITH DISCHARGE PIPING	QUALITROL	XRPD
42	2	3	COMBINATION JACKING PAD, AND LIFTING EYES		
43	AS DES.	BASE	STRUCTURAL BASE		
72	4	COVER	LOW-VOLTAGE BUSHING HOUSINGS		
73	2	1	COMBINATION JACKING PAD, ROCKER, & LIFTING EYES		
74	2	3	LIFTING EYES FOR UPRIGHTING THE TRANSFORMER		
75	1	3	EXTERNAL GAS EXPANSION TANK		

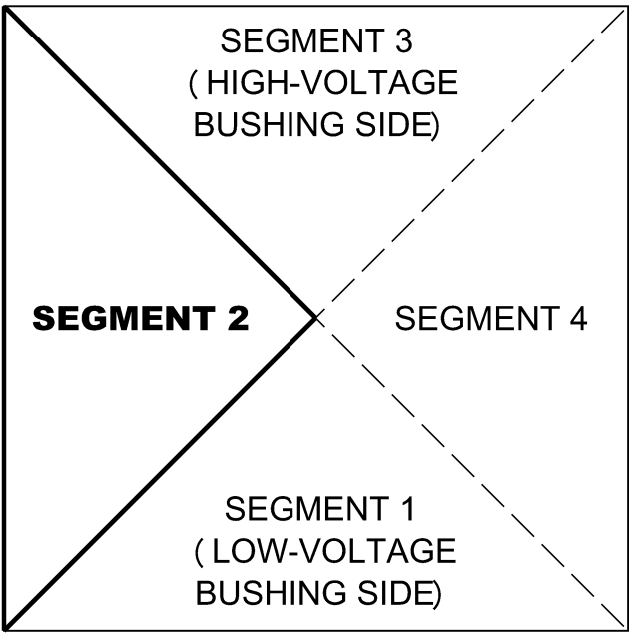


FIGURE 1 - TRANSFORMER TANK SEGMENT IDENTIFICATION (TOP VIEW)

SCALE: NTS



US Army Corps of Engineers®

DATE	DESCRIPTION	MARK
APRIL 2026	UPDATED MAX WEIGHT	A

DESIGNED BY: P. KLEINE, P.E. DRAWN BY: D. FILER CHECKED BY: D. FILER SUBMITTED BY: JORDAN S. FINK, P.E. SIZE: ANSI D	ISSUE DATE: OCTOBER 2025 SOLICITATION NO.: SYSTEM NO.: CONTRACT NO.: FILE NUMBER: LGS-16-1E403 FILENAME: LGS031108a E-103.dgn	U.S. ARMY CORPS OF ENGINEERS WALLA WALLA DISTRICT 201 N 3RD AVE WALLA WALLA, WASHINGTON US ARMY CORPS OF ENGINEERS HYDROELECTRIC DESIGN CENTER 333 SW 1ST AVE PORTLAND, OREGON
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LITTLE GOOSE LOCK AND DAM
LOWER SNAKE RIVER SPARE TRANSFORMERS
T1
GENERATOR STEP-UP TRANSFORMER
LAYOUT AND LIMITING DIMENSIONS
SEGMENT 2

SHEET ID
VOLUME 1
E-103

FINAL

